Annotated Bibliography — Roland Knapp

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About this bibliography

All 81 references in references_cv_knapp.bib are annotated below; 77 summaries are built directly from the published abstract. Summaries are paraphrased, not quoted, and each citation links to the article's DOI where one exists.

Four entries could not be summarized from a published abstract and are flagged in the text:

- **No formal abstract; summarized from the article text (2):** Wilzbach et al. 1988; Knapp, Corn & Schindler 2001.
- **No abstract exists (2):** Land et al. 2026 and Smith et al. 2022 are *Herpetological Review* natural history notes.

Annotated entries

Entries are in chronological order (oldest first), then alphabetical by first author within each year.

Wilzbach, M. A., Cummins, K. W., & Knapp, R. A. (1988). Toward a functional classification of stream invertebrate drift. *Verhandlung Internationale Vereinigung Limnologie*, 23, 1244–1254. https://doi.org/10.1080/03680770.1987.11899798

No formal abstract; summarized from the article text. Wilzbach, Cummins, and Knapp argue that contradictory results in stream drift research stem partly from generalizing across taxa and ignoring species' functional adaptations, and propose a functional classification that distinguishes active from passive drift entry, makes diel timing explicit, separates taxa by susceptibility to dislodgement at high flow, and aligns with existing feeding- and habit-group schemes. Drag rose linearly with current velocity, and the velocity needed to dislodge animals was inversely related to body height-to-width ratio, which the authors propose as an index of dislodgement risk. Streamlined swimmers such as baetid mayflies are likely active drifters, whereas clingers, burrowers, and sprawlers drift mainly through accidental dislodgement. Drifting animals had higher 12-hour mortality than benthic animals, suggesting that passive drifters contribute less to downstream colonization.

Knapp, R. A., & Sargent, R. C. (1989). Egg-mimicry as a mating strategy in the fantail darter, *Etheostoma flabellare*: females prefer males with eggs. *Behavioral Ecology and Sociobiology*, 25, 321–326. https://doi.org/10.1007/BF00302989

Knapp and Sargent tested the hypothesis, proposed by Page and Swofford, that female preference for spawning with males already guarding eggs drove the evolution of egg-mimicking structures in male darters. In fishes with paternal care, this female preference has led males of several species to adopt unrelated eggs as a mating tactic. In choice experiments with fantail darters (*Etheostoma flabellare*), females preferred males with eggs over males without eggs, and also preferred males with egg mimics over males lacking them. The results support the idea that female preference for egg-guarding males can select for specialized egg-mimicking ornaments.

Knapp, R. A., & Dudley, T. L. (1990). Growth and longevity of golden trout, *Oncorhynchus aguabonita*, in their native streams. *California Fish and Game*, 76, 161–173.

Knapp and Dudley aged golden trout (*Oncorhynchus aguabonita*) from 17 streams on the Kern Plateau of the southern Sierra Nevada using otoliths, estimated growth from length–age and weight–age relationships, and related growth, condition, and density to site characteristics with stepwise multiple regression. Populations were highly stunted and generally at low density, yet individuals lived up to 9 years. Growth declined significantly with trout density and increased with aquatic vegetation, bank vegetation, channel stability, and elevation. The authors conclude that although density affects growth, the generally slow growth likely reflects the low productivity of these unstable montane streams and the short high-elevation growing season.

Knapp, R. A., & Warner, R. R. (1991). Male parental care and female choice in the bicolor damselfish, *Stegastes partitus*: bigger is not always better. *Animal Behaviour*, 41, 747–756. https://doi.org/10.1016/S0003-3472(05)80341-0

Knapp and Warner studied female mate choice and egg survival in the bicolor damselfish (*Stegastes partitus*), a coral reef fish with exclusive male parental care. Earlier work on fishes with paternal care had found that females prefer larger males or males guarding larger broods, but in this population neither male size nor brood size affected male mating success. Of the male and territory traits measured, only courtship rate predicted female choice, and egg-hatching success was likewise unrelated to male size or brood size. The authors attribute the absence of preference for large males or broods to patterns of clutch loss and propose testing whether high-courtship males provide better care.

Knapp, R. A., & Kovach, J. T. (1991). Courtship as an honest indicator of male parental quality in the bicolor damselfish, *Stegastes partitus*. *Behavioral Ecology*, 2, 295–300. https://doi.org/10.1093/beheco/2.4.295

Knapp and Kovach tested whether female bicolor damselfish choose mates by courtship rate and whether courtship honestly signals male parental quality. Over two reproductive cycles, male courtship rate was positively correlated with mating success, and males began courting several days before females spawned, giving females time to assess them. Experimental manipulations showed that a male's courtship rate predicted the survival of eggs in his nest. The authors suggest that variation in both courtship and parental ability stems from differences in male energy reserves, and that the results support honest advertising and adaptive models of sexual selection.

Knapp, R. A. (1993). The influence of egg survivorship on the subsequent nest fidelity of female bicolour damselfish, *Stegastes partitus*. *Animal Behaviour*, 46, 111–121. https://doi.org/10.1006/anbe.1993.1166

Knapp tested whether female bicolour damselfish (*Stegastes partitus*), a species with exclusive male parental care, assess nest quality directly from the survival of previously laid eggs rather than only from male traits correlated with quality. Females spawn repeatedly within a reproductive cycle and usually show high nest fidelity, but they abandon nests that lose an entire brood to nocturnally feeding brittlestars. Experimental brood removals and additions of brittlestar odour showed that the odour alone was sufficient to make females avoid a nest, and visual cues of brood loss may also contribute. Because clutches laid in nests after such complete-brood losses were much less likely to survive, avoiding these nests increases the probability that a female's future eggs survive.

Knapp, R. A., Sikkel, P. C., & Vredenburg, V. T. (1995). Age of clutches in nests and the within-nest spawning-site preferences of three damselfish species (Pomacentridae). *Copeia*, 1995, 78–88. https://doi.org/10.2307/1446801

Knapp, Sikkel, and Vredenburg found that females of three damselfish species (*Stegastes partitus*, *S. leucostictus*, and *Hypsypops rubicundus*) choose where within a male's nest to lay eggs, placing new clutches beside early-stage clutches and away from late-stage clutches even when clutches differ in age by only one day. The pattern held in nests with single and with multiple clutches. In *S. partitus*, clutch position had no measurable effect on clutch survival. Of four explanations tested (dilution of egg predation, oxygen competition among clutches, a between-nest preference for early-stage clutches, and a response to male care), the authors judge differential male care of younger clutches the most likely.

Knapp, R. A. (1995). Influence of energy reserves on the expression of a secondary sexual trait in male bicolor damselfish, *Stegastes partitus*. *Bulletin of Marine Science*, 57(3), 672–681.

This study experimentally tested whether energy reserves govern the expression of courtship, a secondary sexual trait, in male bicolor damselfish (*Stegastes partitus*), a species in which males alone care for eggs. Earlier work showed that courtship rate honestly signals male parental quality and is used by females in mate choice, and preliminary data suggested both depend on energy reserves. Males in a natural population whose diets were supplemented courted more than unfed males, and courtship increased with body fat. Unexpectedly, unfed males also reduced courtship over the study, possibly due to changes in natural food availability and male–male interactions, but overall the results show that energy reserves influence secondary sexual trait expression in a marine fish.

Knapp, R. A., & Vredenburg, V. T. (1996). Spawning by California golden trout: characteristics of spawning fish, seasonal and daily timing, redd characteristics, and microhabitat preferences. *Transactions of the American Fisheries Society*, 125(4), 519–531. https://doi.org/10.1577/1548-8659(1996)125<0519:SBCGTC>2.3.CO;2

Knapp and Vredenburg described the spawning biology of California golden trout (*Oncorhynchus mykiss aguabonita*) in the Golden Trout Wilderness, including the influence of stream temperature on seasonal and daily spawning timing, redd characteristics, and the microhabitat chosen by spawning females. Spawning began in mid-May after peak discharge, once daily maximum water temperatures consistently exceeded 15°C, lasted about three weeks, and peaked in the afternoon near the daily temperature maximum. Among mature fish (over 95 mm), males were larger than females and ripe males outnumbered ripe females 3- to 9-fold; early-spawning females were larger and used coarser substrates than later spawners. Females preferred mean substrate sizes of 4–12 mm, depths of 5–20 cm, and velocities of 30–70 cm/s, and golden trout used finer substrates, shallower water, and shallower redds than other salmonids, likely reflecting their unusually small size.

Knapp, R. A., & Vredenburg, V. T. (1996). A field comparison of the substrate composition of California golden trout redds sampled with two devices. *North American Journal of Fisheries Management*, 16, 674–681. https://doi.org/10.1577/1548-8675(1996)016<0674:AFCOTS>2.3.CO;2

Knapp and Vredenburg compared substrate samples from California golden trout redds in Cottonwood Creek, a fine-substrate stream (geometric mean particle diameter about 6 mm), collected with an inexpensive excavated-core sampler (15-cm plastic pipe) and with

a freeze-core sampler. Proportions of individual size fractions and percent fines in the upper excavated cores were similar to those in freeze cores, and the two samplers were equally likely to encounter egg pockets. Geometric mean diameters differed because of differences in particles larger than 12.5 mm, and converged when those were excluded. Power analysis showed generally high power (above 0.8), and the authors conclude that the excavated-core sampler is a viable alternative to freeze coring for redds in small streams with fine substrates.

Knapp, R. A., & Matthews, K. R. (1996). Livestock grazing, golden trout, and streams in the Golden Trout Wilderness, California: impacts and management implications. *North American Journal of Fisheries Management*, 16, 805–820. https://doi.org/10.1577/1548-8675(1996)016<0805:LGGTAS>2.3.CO;2

Knapp and Matthews compared stream habitat and California golden trout populations inside and outside livestock exclosures in the Golden Trout Wilderness over two consecutive years. Most physical stream characteristics differed substantially between grazed and ungrazed areas, consistent with recovery inside exclosures: ungrazed areas had more canopy shade, deeper and narrower channels, and higher bankfull heights. Golden trout densities and biomass at the study sites were among the highest recorded for stream-dwelling trout in the western United States, and in three comparisons were significantly higher per unit area in ungrazed reaches, though differences per unit stream length were less consistent. The authors conclude that existing grazing levels were degrading stream and riparian habitat to the detriment of golden trout.

Knapp, R. A., Vredenburg, V. T., & Matthews, K. R. (1998). Effects of stream channel morphology on golden trout spawning habitat and recruitment. *Ecological Applications*, 8(4), 1104–1117. https://doi.org/10.1890/1051-0761(1998)008[1104:EOSCMO]2.0.CO;2

Knapp, Vredenburg, and Matthews tested whether stream channel morphology controls spawning habitat availability for California golden trout and whether that habitat limits age-0 production and recruitment, challenging the prevailing paradigm of strong density-dependent mortality in stream salmonids. Wide stream reaches contained more spawning habitat, more redds, and higher age-0 densities than narrow reaches, and age-0 mortality was largely density-independent. Cohort densities were positively correlated across years for ages 0–2, indicating that spawning habitat limited both age-0 production and adult recruitment. Because cattle grazing had widened channels, grazed sections supported more spawning habitat and higher trout densities, but negatively density-dependent growth meant those densities likely came at the cost of slower growth.

Knapp, R. A., & Matthews, K. R. (1998). Eradication of nonnative fish by gill netting from a small mountain lake in California. *Restoration Ecology*, 6(2), 207–213. https://doi.org/10.1111/j.1526-100X.1998.06212.x

Knapp and Matthews tested intensive gill netting as a non-toxic alternative to rotenone for eradicating introduced trout from a small subalpine lake in the Sierra Nevada. They removed both the resident trout population and a second population that was accidentally stocked during the project, using 18 and 15 net sets, respectively. Adult trout were easily captured, but juveniles were rarely caught until they reached about 110 mm. Based on morphometry of 330 Sierra Nevada lakes, the authors estimated that gill netting could succeed in 15–20% of lakes, mainly small, isolated lakes with limited spawning habitat, making it a slower and costlier but viable alternative under some conditions.

Knapp, R. A., & Preisler, H. K. (1999). Is it possible to predict habitat use by spawning salmonids? A test using California golden trout (*Oncorhynchus mykiss aguabonita*). *Canadian Journal of Fisheries and Aquatic Sciences*, 56(9), 1576–1584. https://doi.org/10.1139/f99-081

Knapp and Preisler tested whether salmonid spawning sites can be predicted from habitat features, a widely held assumption that earlier attempts had struggled to support. Using California golden trout (*Oncorhynchus mykiss aguabonita*), they applied nonparametric logistic regression to identify habitat features associated with spawning locations, then fit a parametric model to predict the probability that particular stream locations would be used. The nonparametric model explained 62% of the variation in spawning location, and four of eight habitat variables (substrate size, water depth, water velocity, and stream width) together explained 59%; cross-validation indicated good fit. The authors conclude that spawning-site use can be predicted from easily measured habitat variables and advocate wider use of nonparametric regression in fish-habitat modeling.

Knapp, R. A., & Matthews, K. R. (2000). Non-native fish introductions and the decline of the mountain yellow-legged frog from within protected areas. *Conservation Biology*, 14(2), 428–438. https://doi.org/10.1046/j.1523-1739.2000.99099.x

Knapp and Matthews examined whether introduced trout contributed to the decline of the mountain yellow-legged frog (*Rana muscosa*) within protected areas of the Sierra Nevada that were presumed to be undisturbed. They surveyed more than 1,700 sites in two adjacent, historically fishless protected areas that differed mainly in the extent of fish stocking. Fish negatively affected frogs at the landscape, watershed, and individual water-body scales; after accounting for habitat, frogs were three times more likely to occur and six times more abundant in fishless water bodies. Because frog larvae depend

on the same deep water bodies where fish were stocked, and fish can be removed from some lakes with modest effort, the authors suggest the decline may be relatively easy to reverse.

Knapp, R. A., Matthews, K. R., & Sarnelle, O. (2001). Resistance and resilience of alpine lake fauna to fish introductions. *Ecological Monographs*, 71(3), 401–421. https://doi.org/10.1890/0012-9615(2001)071[0401:RAROAL]2.0.CO;2

Knapp, Matthews, and Sarnelle quantified the resistance and resilience of amphibians, benthic macroinvertebrates, and zooplankton in naturally fishless Sierra Nevada alpine lakes to trout introductions, comparing never-stocked lakes, stocked lakes still containing fish, and formerly stocked lakes that had been fishless for 5–10, 11–20, or more than 20 years. The fauna had low resistance: mountain yellow-legged frogs, conspicuous benthic macroinvertebrates, and large crustacean zooplankton were sharply reduced by fish, whereas small and inconspicuous taxa were unaffected or increased. Resilience was generally high, with both macroinvertebrate and zooplankton communities converging on never-stocked conditions 11–20 years after fish disappeared, aided by winged adult insects, zooplankton resting eggs, and nearby frog source populations. The authors caution that resilience may now be lower because many frog populations have since disappeared and zooplankton egg banks lose viability over time.

Knapp, R. A., Garton, J. A., & Sarnelle, O. (2001). The use of egg shells to infer the historical presence of copepods in alpine lakes. *Journal of Paleolimnology*, 25(4), 539–543. https://doi.org/10.1023/A:1011178406716

Knapp, Garton, and Sarnelle note that copepods are rarely used in lake paleoecology because they lack durable exoskeletons, and they evaluated eggshells of *Hesperodiaptomus* copepods, which are abundant and well preserved in alpine lake sediments, as an alternative. Sediment eggshells matched laboratory-produced eggshells in size and shape, and diapausing eggs recovered from sediments hatched into nauplii indistinguishable from those of laboratory *Hesperodiaptomus*. Eggshells were roughly two orders of magnitude more abundant in cores than viable eggs, making them a better indicator of historical presence. The method can identify lakes where fish introductions eliminated *Hesperodiaptomus* and where reintroduction may be needed after fish removal.

Knapp, R. A., Corn, P. S., & Schindler, D. E. (2001). The introduction of nonnative fish into wilderness lakes: good intentions, conflicting mandates, and unintended consequences. *Ecosystems*, 4(4), 275–278. https://doi.org/10.1007/s10021-001-0009-0

No formal abstract; summarized from the article text. Knapp, Corn, and Schindler introduce an *Ecosystems* special feature, arising from a 1998 workshop of managers and scientists, on stocking nonnative trout into historically fishless lakes in protected areas of western North America. About 60% of the roughly 16,000 naturally fishless western lakes, and 95% of larger, deeper ones, now contain trout, and more than 7,000 lakes are still regularly stocked, often by aircraft, mainly in wilderness areas. They note that earlier research focused narrowly on direct effects in single lakes, and summarize the feature's papers on policy and jurisdiction, downstream invasion, phosphorus cycling and algal production, landscape effects on amphibians, and incomplete recovery after fish removal. They conclude that trout effects propagate through food webs and across landscapes, so managers must evaluate stocking at ecosystem and landscape scales.

Matthews, K. R., Pope, K. L., Preisler, H. K., & Knapp, R. A. (2001). Effects of nonnative trout on Pacific treefrogs (*Hyla regilla*) in the Sierra Nevada. *Copeia*, 101(4), 1130–1137. https://doi.org/10.1643/0045-8511(2001)001[1130:EONTOP]2.0.CO;2

Matthews, Pope, Preisler, and Knapp surveyed about 1,700 water bodies across roughly 100,000 ha of the John Muir Wilderness and Kings Canyon National Park to evaluate how nonnative trout affect Pacific treefrogs (*Hyla regilla*) in the High Sierra. At the landscape scale, 7.2% of water bodies in trout-rich John Muir Wilderness contained frogs versus 26.6% in Kings Canyon, where the proportion of water surface area occupied by frogs was 19.4 times higher. At the water-body scale, after accounting for habitat and isolation, the odds of frog presence were 2.4 times higher in fishless water bodies, and expected frog numbers were 3.7 times higher where trout were absent; frogs were also more likely at lower elevations and in shallow, silty water bodies. The authors conclude that treefrogs have likely declined where trout are common, adding to evidence of landscape-scale impacts of introduced fish on native species.

Schindler, D. E., Knapp, R. A., & Leavitt, P. R. (2001). Alteration of nutrient cycles and algal production resulting from fish introductions into mountain lakes. *Ecosystems*, 4(4), 308–321. https://doi.org/10.1007/s10021-001-0013-4

Schindler, Knapp, and Leavitt examined how introducing salmonids into naturally fishless mountain lakes alters nutrient cycling and primary production. Bioenergetics modeling in Sierra Nevada lakes showed that phosphorus recycled by trout increased with fish biomass rather than density, that lakes maintained by stocking recycled phosphorus at more than twice the rate of lakes with naturally reproducing fish, and that fish-regenerated phosphorus roughly equaled atmospheric phosphorus deposition. Paleolimnological analyses of lakes in the southern Canadian Rockies showed that algal production rose substantially after trout introduction and stayed elevated while fish

persisted. The authors conclude that trout boost primary production by making benthic phosphorus available to the pelagic zone, creating a management tension between recreational fisheries and natural ecosystem processes.

Matthews, K. R., Knapp, R. A., & Pope, K. L. (2002). Garter snake distributions in high-elevation aquatic ecosystems: is there a link with declining amphibian populations and nonnative trout introductions? *Journal of Herpetology*, 36(1), 16–22. https://doi.org/10.1670/0022-1511(2002)036[0016:GSDIHE]2.0.CO;2

Matthews, Knapp, and Pope tested whether amphibian declines caused by trout introductions in the Sierra Nevada have, in turn, reduced populations of the mountain garter snake (*Thamnophis elegans elegans*), an amphibian specialist. Across 2,103 high-elevation lakes, snakes were 30 times more likely to occur in lakes with amphibians, and lakes with snakes had about six times as many amphibians within 1 km as lakes without. In Kings Canyon National Park, where 40% of larger lakes contain trout, amphibians occurred in 52% of lakes and snakes in 33 of 1,059 lakes; in the John Muir Wilderness, where 80% of larger lakes contain trout, amphibians occurred in 19% of lakes and no snakes were found in 1,044 lakes. The authors conclude that trout introductions have likely caused declines in both amphibians and their snake predators.

Knapp, R. A., Matthews, K. R., Preisler, H. K., & Jellison, R. (2003). Developing probabilistic models to predict amphibian site occupancy in a patchy landscape. *Ecological Applications*, 13(4), 1069–1082. https://doi.org/10.1890/1051-0761(2003)13[1069:DPMTPA]2.0.CO;2

Knapp, Matthews, Preisler, and Jellison developed probabilistic models of patch occupancy for the mountain yellow-legged frog (*Rana muscosa*), whose habitat has been fragmented by the introduction of predatory fish. A semiparametric logistic regression with habitat variables, fish presence, and a spatial location term fit well but could only be applied within the study area, so the authors also built a more general autologistic model using an autocovariate measuring each lake's isolation from neighboring frog populations. After accounting for spatial autocorrelation, both models showed that frog presence depended strongly on lake depth, elevation, fish presence, substrate, and isolation. Cross-validation favored the autologistic model for prediction, which the authors suggest could be used to assess the likelihood of population persistence.

Armstrong, T. W., & Knapp, R. A. (2004). Response by trout populations in alpine lakes to an experimental halt to stocking. *Canadian Journal of Fisheries and Aquatic Sciences*, 61(11), 2025–2037. https://doi.org/10.1139/f04-144

Armstrong and Knapp tested the rarely examined assumption that trout populations in stocked alpine lakes would disappear without continued stocking. They surveyed 95 lakes in the John Muir Wilderness before and after a 4–8-year experimental halt to stocking, and 84 lakes in Sequoia–Kings Canyon National Park after a much longer stocking hiatus. Based on recruitment during the no-stocking periods, 70% and 68% of lakes, respectively, contained naturally reproducing trout, and lakes with at least 2.1 m² of spawning habitat below roughly 3,520 m elevation nearly always supported reproduction. Halting stocking did not significantly change trout density in reproducing populations and may have increased individual growth, leading the authors to conclude that most stocking of Sierra Nevada alpine lakes could be permanently halted with minimal effect on recreational fishing.

Sarnelle, O., & Knapp, R. A. (2004). Zooplankton recovery after fish removal: Limitations of the egg bank. *Limnology and Oceanography*, 49(4/2), 1382–1392. https://doi.org/10.4319/lo.2004.49.4_part_2.1382

Sarnelle and Knapp removed fish from four Sierra Nevada alpine lakes that had contained both the calanoid copepod *Hesperodiaptomus shoshone* and the cladoceran *Daphnia middendorfiana* before trout stocking eliminated them. *D. middendorfiana*, which can reproduce asexually, returned to all four lakes, whereas the obligately sexual *H. shoshone* returned to none, despite diapausing eggs in two lakes' sediments and a few individuals appearing in one lake. The founding population implied for *D. middendorfiana* in one lake, where it took four years to become detectable, was three to four orders of magnitude smaller than the minimum founding population calculated for *H. shoshone*, suggesting a strong Allee effect. Combined with low overland dispersal, mate limitation means *H. shoshone* may never recolonize these lakes without reintroduction.

Adams, M. J., Hossack, B. R., Knapp, R. A., Corn, P. S., Diamond, S. A., Trenham, P. C., & Fagre, D. B. (2005). Distribution patterns of lentic-breeding amphibians in relation to ultraviolet radiation exposure in western North America. *Ecosystems*, 8(5), 488–500. https://doi.org/10.1007/s10021-003-0033-3

Adams and colleagues tested whether ultraviolet-B (UV-B) radiation, a proposed driver of amphibian declines, explains where pond- and lake-breeding amphibians occur in western North America. They surveyed 683 water bodies in Glacier, Olympic, and Sequoia–Kings Canyon national parks and estimated UV-B exposure from elevation, surrounding topography, and dissolved organic carbon. Only two of eight species showed occupancy patterns consistent with UV-B harm, with the strongest evidence for the long-toed salamander (*Ambystoma macrodactylum*) in Glacier; in Olympic that species' occurrence

actually increased with UV-B. Most species showed no UV-B effect, and the authors conclude that UV-B is unlikely to be a widespread cause of amphibian loss on its own, though interactions with other stressors deserve further study.

Briggs, C. J., Vredenburg, V. T., Knapp, R. A., & Rachowicz, L. J. (2005). Investigating the population-level effects of chytridiomycosis: an emerging infectious disease of amphibians. *Ecology*, 86(12), 3149–3159. https://doi.org/10.1890/04-1428

Briggs and colleagues built simple within-lake models of chytridiomycosis dynamics to test whether existing knowledge of the disease can explain field observations of mountain yellow-legged frogs in the Sierra Nevada. Those observations include local extinctions across a wide range of host population sizes alongside some populations that persist despite heavily infected tadpoles. Although the disease killed every infected post-metamorphic frog in laboratory trials, the models showed that long-term persistence can only occur if at least some infected adults survive and reproduce. The paper thus identifies adult survival with infection as the key unknown governing population-level outcomes.

Brooks, P. D., O'Reilly, C. M., Diamond, S. A., Campbell, D. H., Knapp, R., Bradford, D., Corn, P. S., Hossack, B., & Tonnessen, K. (2005). Spatial and temporal variability in the amount and source of dissolved organic carbon: implications for ultraviolet exposure in amphibian habitats. *Ecosystems*, 8(5), 478–487. https://doi.org/10.1007/s10021-003-0031-5

Brooks and colleagues measured the concentration, chemistry, and sources of dissolved organic carbon (DOC) and the resulting attenuation of UV-B radiation at four Rocky Mountain National Park sites from 1999 to 2001, adding eight sites in California and Montana parks in 2000. UV-B attenuation was significantly but only modestly related to DOC concentration because the amount of UV-B absorbed per unit DOC varied over time and among sites. Fluorescence analysis of fulvic acids showed that DOC source mattered: California sites were dominated by algal-derived DOC, which allowed much deeper UV-B penetration than terrestrially derived DOC. Interactions among vegetation, soils, precipitation, and hydrology made UV-B exposure in amphibian habitats hard to predict from DOC concentration alone.

Brose, U., Cushing, L., Berlow, E. L., Jonsson, T., Banasek-Richter, C., Bersier, L., Blanchard, J. L., Brey, T., Carpenter, S. R., Blandenier, M. C., Cohen, J. E., Dawah, H. A., Dell, T., Edwards, F., Harper-Smith, S., Jacob, U., Knapp, R. A., Ledger, M. E., Memmott, J., Mintenbeck, K., Pinnegar, J. K., Rall, B. C., Rayner, T., Ruess, L., Ulrich, W., Warren, P., Williams, R. J., Woodward, G., Yodzis, P., & Martinez, N. D. (2005). Body sizes of consumers and their resources. *Ecology*, 86(9), 2545. https://doi.org/10.1890/05-0379

Brose and a large group of coauthors compiled a data set of consumer and resource body sizes drawn from food webs in freshwater, marine, and terrestrial ecosystems, including predator–prey and parasitoid–host interactions. With 16,807 records, it was described as the largest such compilation assembled to that point. Each record includes body sizes along with taxonomy, location, habitat, feeding type, and metabolic category of both partners. The data set was designed to support research on how body-size relationships shape food-web structure, interaction strengths, and population stability.

Diamond, S. A., Trenham, P. C., Adams, M. J., Hossack, B. R., Knapp, R. A., Stark, S. L., Bradford, D., Corn, P. S., Czarnowski, K., Brooks, P. D., Fagre, D., Breen, B., Detenbeck, N. E., & Tonnessen, K. (2005). Estimated ultraviolet radiation doses in wetlands in six national parks. *Ecosystems*, 8(5), 462–477. https://doi.org/10.1007/s10021-003-0030-6

Diamond and colleagues estimated ultraviolet-B doses at 1 cm depth for 1,024 wetlands in six U.S. national parks (Acadia, Glacier, Great Smoky Mountains, Olympic, Rocky Mountain, and Sequoia–Kings Canyon) by combining ground-based UV-B measurements, solar radiation models, GIS terrain and vegetation data, and measurements of dissolved organic carbon (DOC). Summer-solstice doses averaged 19.3 W·h/m² and differed significantly among all parks, from lowest in Acadia to highest in Rocky Mountain. DOC reduced potential maximum doses by 1–87% and landscape shading by 0–77%, with Sequoia having the lowest DOC and least shading. The authors conclude that assessing UV-B effects on aquatic organisms requires accounting for all components of exposure.

Harper-Smith, S., Berlow, E. L., Knapp, R. A., Williams, R. J., & Martinez, N. D. (2005). Communicating ecology through food webs: visualizing and quantifying the effects of stocking alpine lakes with trout. In P. C. de Ruiter, V. Wolters, & J. C. Moore (Eds.), *Dynamic food webs - multispecies assemblages, ecosystem development and environmental change* (pp. 407–423). Academic Press.

This book chapter presents food webs as an intuitive way to communicate the system-wide effects of human activities to resource managers, using trout stocking in alpine lakes as a case study. The authors note that food webs are common in teaching but rarely used in management, where single-species approaches still dominate and feeding relationships are seldom considered. Simple binary food webs compactly illustrated the large changes in alpine lake communities following trout introduction and their recovery after fish removal, with stocked lakes having much simpler webs than never-stocked lakes. The authors caution that link structure alone cannot capture species dynamics, which also depend on recruitment, abiotic conditions, disturbance, and non-trophic interactions.

Knapp, R. A., Hawkins, C. P., Ladau, J., & McClory, J. G. (2005). Fauna of Yosemite National Park lakes has low resistance but high resilience to fish introductions. *Ecological Applications*, 15(3), 835–847. https://doi.org/10.1890/04-0619

Knapp and colleagues used the ratio of observed to expected taxa (O/E) to compare amphibians, reptiles, benthic macroinvertebrates, and zooplankton among never-stocked lakes, stocked lakes still containing fish, and stocked lakes that had reverted to fishless conditions in Yosemite National Park. Lakes with fish had 16% fewer taxa than never-stocked lakes (O/E 0.84 vs. 1.00), and resistance declined with fish density and elevation but increased with depth. Formerly stocked lakes that had become fishless had an O/E of 1.00, indistinguishable from never-stocked lakes, and their composition resembled never-stocked lakes within two years. The fauna therefore showed low resistance but high resilience, recovering quickly and predictably after fish disappeared.

Knapp, R. A. (2005). Effects of nonnative fish and habitat characteristics on lentic herpetofauna in Yosemite National Park, USA. *Biological Conservation*, 121(2), 265–279. https://doi.org/10.1016/j.biocon.2004.05.003

Knapp used census data from all 2,655 lentic water bodies in Yosemite National Park and generalized additive models to quantify how nonnative trout and habitat characteristics affect the distributions of four amphibian and two reptile species. All species were associated with habitat, particularly elevation and water depth. The mountain yellow-legged frog and Pacific treefrog were strongly negatively associated with trout, whereas the Yosemite toad and Sierra newt were not; the mountain garter snake was negatively associated with trout and positively associated with native amphibians, with a similar but weaker pattern for the Sierra garter snake. The author concludes that trout introductions have substantially altered Yosemite's herpetofauna and that long-term studies are needed to determine whether trout removal can reverse these effects.

Sarnelle, O., & Knapp, R. A. (2005). Nutrient recycling by fish versus zooplankton grazing as drivers of the trophic cascade in alpine lakes. *Limnology and Oceanography*, 50(6), 2032–2042. https://doi.org/10.4319/lo.2005.50.6.2032

Sarnelle and Knapp used whole-lake fish removals from small Sierra Nevada alpine lakes to compare the effects of nutrient excretion by zooplanktivorous fish and grazing by herbivorous zooplankton on phytoplankton biomass and phosphorus (P) limitation. During 1–6 years after removal, while herbivorous zooplankton were unchanged, phytoplankton biomass and the intensity of P limitation changed little; both declined significantly only after *Daphnia* re-established. Fish removal also sharply increased benthic macroinvertebrate biomass, but its timing indicated that benthic P recycling did not confound the comparison, and budgets of P recycled by fish and zooplankton relative

to phytoplankton P demand also showed fish recycling to be a minor P source. The authors conclude that, relative to *Daphnia* grazing, P recycling by fish is relatively unimportant as a driver of trophic cascades in these lakes.

Knapp, R. A., & Morgan, J. A. T. (2006). Tadpole mouthpart depigmentation as an accurate indicator of chytridiomycosis, an emerging disease of amphibians. *Copeia*, 2006(2), 188–197. https://doi.org/10.1643/0045-8511(2006)6[188:TMDAAA]2.0.CO;2

Knapp and Morgan evaluated whether visually scoring mouthpart depigmentation in mountain yellow-legged frog (*Rana muscosa*) tadpoles can accurately diagnose chytridiomycosis, using PCR assays to confirm infection status. Depigmentation was associated with infection, most strongly on the upper jaw sheaths. Classifying tadpoles with fully pigmented upper jaw sheaths as uninfected and all others as infected correctly classified 86% of individuals, and averaging jaw-sheath scores across tadpoles at each site correctly categorized 92% of populations. The authors note that similar studies are needed in other anuran species to determine how broadly the method applies.

Rachowicz, L. J., Knapp, R. A., Morgan, J. A. T., Stice, M. J., Vredenburg, V. T., Parker, J. M., & Briggs, C. J. (2006). Emerging infectious disease as a proximate cause of amphibian mass mortality. *Ecology*, 87(7), 1671–1683. https://doi.org/10.1890/0012-9658(2006)87[1671:EIDAAP]2.0.CO;2

Rachowicz and colleagues evaluated chytridiomycosis, caused by *Batrachochytrium dendrobatidis*, as a cause of declines of the mountain yellow-legged frog (*Rana muscosa*) in the southern Sierra Nevada by combining repeat surveys of several hundred populations with laboratory and field experiments. Mouthpart inspections showed signs of infection in 19% of populations in both 2003 (144 populations) and 2004 (132 populations), and 16% of populations uninfected in 2003 were infected by 2004. Populations declined by an average of 88% following outbreaks at six sites, while populations at seven uninfected sites increased by 45% over the same period. In the laboratory, all infected frogs developed fatal chytridiomycosis after metamorphosis, and in field enclosures 96% of metamorphs at infected sites died versus 5% at uninfected sites, establishing the disease as the proximate cause of many declines.

Davidson, C., & Knapp, R. A. (2007). Multiple stressors and amphibian declines: Dual impacts of pesticides and fish on yellow-legged frogs. *Ecological Applications*, 17(2), 587–597. https://doi.org/10.1890/06-0181

Davidson and Knapp evaluated the combined effects of introduced fish and windborne agricultural pesticides on the mountain yellow-legged frog, the first study to assess both stressors together. They surveyed 6,831 Sierra Nevada water bodies for frogs, fish, and

habitat characteristics and estimated pesticide exposure from upwind application records. Generalized additive models showed that, after accounting for habitat, both fish and pesticides significantly reduced the probability of frog presence, with the landscape-scale pesticide effect considerably stronger. The authors argue that amphibian declines often have multiple causes and that single-factor studies should not be used to dismiss other stressors.

Fisk, D. L., Latta, L. C., Knapp, R. A., & Pfrender, M. E. (2007). Rapid evolution in response to introduced predators I: rates and patterns of morphological and life-history trait divergence. *BMC Evolutionary Biology*, 7(1), 22. https://doi.org/10.1186/1471-2148-7-22

Fisk and colleagues compared *Daphnia melanica* from Sierra Nevada alpine lakes containing introduced salmonids with populations from lakes that never held fish to measure the pace of evolutionary response to a novel predator. *Daphnia* in fish-containing lakes evolved rapid, adaptive reductions in body size and earlier reproduction, with size-related traits declining by up to 13%. Estimated evolutionary rates reached 4,238 darwins (0.036 haldanes), which the authors compare with published rates across ecological timescales. The study shows that species introductions can quickly reshape the selective environment of native species, with long-term consequences for community structure.

Knapp, R. A., Boiano, D. M., & Vredenburg, V. T. (2007). Removal of nonnative fish results in population expansion of a declining amphibian (mountain yellow-legged frog, *Rana muscosa*). *Biological Conservation*, 135(1), 11–20. https://doi.org/10.1016/j.biocon.2006.09.013

Knapp, Boiano, and Vredenburg tested whether removing introduced trout from three Sierra Nevada lakes in two watersheds would allow small remnant populations of the mountain yellow-legged frog (*Rana muscosa*) to recover, and whether recovering populations would begin colonizing nearby habitat. Frog densities increased significantly at all three lakes, and these increases were significantly greater than at fishless control lakes monitored over the same period. As populations grew, frogs dispersed 200–900 m along streams or overland to adjacent unoccupied habitat, the beginning of a metapopulation structure. The authors conclude that large-scale fish removal could at least partly reverse the species' decline, but stress the need for long-term monitoring given the emerging threat of chytridiomycosis.

Latta, L. C., Bakelar, J. W., Knapp, R. A., & Pfrender, M. E. (2007). Rapid evolution in response to introduced predators II: the contribution of adaptive plasticity. *BMC Evolutionary Biology*, 7(1), 21. https://doi.org/10.1186/1471-2148-7-21

Latta and colleagues used common-garden experiments with and without fish chemical cues (kairomones) to separate evolutionary and plastic contributions to trait changes in *Daphnia melanica* from Sierra Nevada lakes with differing fish-stocking histories. Fish introduction was associated with smaller body size and lower egg number, changes that had a genetic basis but were partly induced by fish cues. Body size shifted by about nine phenotypic standard deviations, of which adaptive plasticity explained roughly 11%, while reproductive timing showed no change. The authors conclude that initial plasticity followed by genetic adaptation produced the observed divergence, consistent with Baldwin/Bogert effects.

Morgan, J. A. T., Vredenburg, V. T., Rachowicz, L. J., Knapp, R. A., Stice, M. J., Tunstall, T., Bingham, R. E., Parker, J. M., Longcore, J. E., Moritz, C., Briggs, C. J., & Taylor, J. W. (2007). Population genetics of the frog-killing fungus *Batrachochytrium dendrobatidis*. *Proceedings of the National Academy of Sciences, USA*, 104(34), 13845–13850. https://doi.org/10.1073/pnas.0701838104

Morgan and colleagues studied the population genetics of the amphibian chytrid fungus *Batrachochytrium dendrobatidis*, comparing isolates from the Sierra Nevada of California, a region of intensively studied frog declines. Several lines of evidence pointed to a novel, recently spread pathogen: low genetic diversity, no host specificity, weak association between fungal genotype and geography, a single genotype responsible for local frog extinctions, and indications of human-assisted dispersal. At local scales, however, some fungal populations were genetically diverse and showed evidence of recombination, more consistent with endemism. The authors conclude that neither epidemic spread nor endemism alone explains the decline, and that recombination could generate resistant life stages that aid spread and persistence, complicating global control.

Vredenburg, V. T., Bingham, R., Knapp, R., Morgan, J. A. T., Moritz, C., & Wake, D. (2007). Concordant molecular and phenotypic data delineate new taxonomy and conservation priorities for the endangered mountain yellow-legged frog. *Journal of Zoology*, 271(4), 361–374. https://doi.org/10.1111/j.1469-7998.2006.00258.x

Vredenburg and colleagues resolved the taxonomy of the mountain yellow-legged frog, once abundant in the Sierra Nevada and the Transverse Ranges of southern California but now severely reduced despite largely protected habitat. Bayesian analysis of 1,901 base pairs of mitochondrial DNA revealed two deeply divergent lineages that nearly meet in the Sierra Nevada and are also distinguishable by morphology and advertisement calls, leading the authors to recognize *Rana sierrae* (northern and central Sierra Nevada) and *R. muscosa* (southern Sierra Nevada and southern California). The existing listed population segment did not match these species boundaries. Resurveys of 225 historic localities

showed that 93.3% of *R. sierrae* and 95.2% of *R. muscosa* populations were extinct, and the authors recommend extending protected status to both species across their full ranges.

Knapp, R. A., & Sarnelle, O. (2008). Recovery after local extinction: factors affecting re-establishment of alpine lake zooplankton. *Ecological Applications*, 18(8), 1850–1859. https://doi.org/10.1890/08-0044.1

Knapp and Sarnelle asked why two large alpine zooplankton species, the cladoceran *Daphnia melanica* and the copepod *Hesperodiaptomus shoshone*, do or do not recover after trout disappear, sampling zooplankton and sediments from 44 naturally fishless Sierra Nevada lakes stocked for varying lengths of time. They predicted that asexually reproducing *D. melanica* would recover more often than obligately sexual *H. shoshone*, and that recovery would decline with fish residence time and increase with lake depth because of their effects on egg banks. *D. melanica* recovered with a probability of 0.82 versus 0.54 for *H. shoshone*; recovery of *H. shoshone* depended on lake depth, fish residence time, and elevation, whereas only elevation mattered for *D. melanica*. The authors attribute the difference to reproductive mode and the much longer viability of *Daphnia* resting eggs, and suggest that *H. shoshone* often fails to recover where fish were present for more than 50 years.

Kramer, A. M., Sarnelle, O., & Knapp, R. A. (2008). Allee effect limits colonization success of sexually reproducing zooplankton. *Ecology*, 89(10), 2760–2769. https://doi.org/10.1890/07-1505.1

Kramer, Sarnelle, and Knapp used fish introductions and removals in Sierra Nevada alpine lakes as whole-ecosystem experiments to test whether an Allee effect limits recovery of the copepod *Hesperodiaptomus shoshone*, which often fails to return after fish disappear despite a long-lived egg bank. They hypothesized that population growth in this dioecious species is limited by mate encounter rates and estimated a critical density of 0.5–5 copepods/m³. When copepods were stocked into lakes across that range, only the lake with the highest initial density (3 copepods/m³) recovered, even though caged copepods survived and reproduced normally, confirming habitat suitability. Mating success increased with density in experimental and natural populations, and a mesocosm study of another diaptomid showed similar density-dependent growth, supporting mate limitation as the mechanism.

Briggs, C. J., Knapp, R. A., & Vredenburg, V. T. (2010). Enzootic and epizootic dynamics of the chytrid fungal pathogen of amphibians. *Proceedings of the National Academy of Sciences, USA*, 107(21), 9695–9700. https://doi.org/10.1073/pnas.0912886107

Briggs, Knapp, and Vredenburg asked why the amphibian chytrid fungus (*Batrachochytrium dendrobatidis*, Bd) drives some Sierra Nevada frog populations extinct while others decline and then persist at low density for years. A five-year field study showed that adults in persistent populations carried low fungal loads, survived between years, and frequently lost and regained infection. Their model shows that both outcomes can emerge from density-dependent infection dynamics in the same host–pathogen system, without invoking differences in host susceptibility, pathogen virulence, or environment. The long-lived tadpole stage and non-host Bd reservoirs help the pathogen persist in enzootic populations.

Epanchin, P. N., Knapp, R. A., & Lawler, S. P. (2010). Nonnative trout impact an alpine-nesting bird by altering aquatic-insect subsidies. *Ecology*, 91(8), 2406–2415. https://doi.org/10.1890/09-1974.1

Epanchin, Knapp, and Lawler examined how introduced trout in Sierra Nevada lakes indirectly affect gray-crowned rosy-finches (*Leucosticte tephrocotis dawsoni*), which forage on aquatic insects emerging from alpine lakes. Trout competed with the finches for emerging mayflies, which were 98% less abundant in lakes containing fish than in fishless lakes. Rosy-finches concentrated around insect emergence and were 5.9 times more abundant at fishless lakes. Because emergence coincides with nestling provisioning, trout may have reduced rosy-finch populations below historical levels, and the authors recommend including subsidy-dependent terrestrial species in alpine lake restoration planning.

Latta, L. C., Fisk, D. L., Knapp, R. A., & Pfrender, M. E. (2010). Genetic resilience of *Daphnia* populations following experimental removal of introduced fish. *Conservation Genetics*, 11(5), 1737–1745. https://doi.org/10.1007/s10592-010-0067-y

Latta, Fisk, Knapp, and Pfrender examined how quickly genetic diversity recovers in alpine zooplankton after local extinction, studying two Sierra Nevada populations of *Daphnia melanica* that introduced salmonids had extirpated and that re-established after experimental fish removal. Using nuclear and mitochondrial markers and a common-garden experiment for quantitative genetic variation, they found that nuclear diversity in the first year after fish removal was only slightly lower than in nearby fishless populations. Within eight years, the restored populations held substantial mitochondrial and quantitative genetic diversity, likely derived from immigration and from hatching of locally stored diapausing eggs. The authors conclude that organisms with dormant egg banks and asexual reproduction can be genetically resilient to demographic collapse.

Vredenburg, V. T., Knapp, R. A., Tunstall, T. S., & Briggs, C. J. (2010). Dynamics of an emerging disease drive large-scale amphibian population extinctions. *Proceedings of the National Academy of Sciences, USA*, 107(21), 9689–9694. https://doi.org/10.1073/pnas.0914111107

Vredenburg, Knapp, Tunstall, and Briggs used fine-scale data on the arrival and spread of *Batrachochytrium dendrobatidis* (Bd) through three Sierra Nevada lake basins, each with multiple mountain yellow-legged frog populations, to explain how a pathogen can drive hosts to extinction despite theory predicting density-dependent fade-out. Bd was undetectable on frogs until just before epizootics began, then spread between populations as a wave at roughly 700 m per year. Within populations, prevalence rapidly reached 100% and infection intensity rose synchronously, with mass mortality beginning only after intensity crossed a critical threshold, repeatedly leading to extinction. Because Bd's high growth rate and virulence allowed it to infect nearly all hosts and reach lethal loads before density-dependent effects could act, the authors suggest that keeping infection intensity below the threshold could prevent extinctions.

Bradford, D. F., Knapp, R. A., Sparling, D. W., Nash, M. S., Stanley, K. A., Tallent-Halsell, N. G., McConnell, L. L., & Simonich, S. M. (2011). Pesticide distributions and population declines of California, USA, alpine frogs, *Rana muscosa* and *Rana sierrae*. *Environmental Toxicology and Chemistry*, 30(3), 682–691. https://doi.org/10.1002/etc.425

Bradford and colleagues directly tested the hypothesis that pesticides drifting from California's Central Valley caused declines of the mountain yellow-legged frogs *Rana muscosa* and *R. sierrae*, which earlier studies had evaluated mainly with upwind pesticide-use estimates or sparse measurements. They measured pesticides in air, sediment, and Sierran treefrog (*Pseudacris sierra*) tadpoles, plus tadpole cholinesterase activity, at 28 high-elevation sites in the southern Sierra Nevada, each sampled twice. Nine of 46 compounds analyzed were detected in 30% of samples, but frog population status showed no negative relationship with any pesticide concentration or with cholinesterase activity. Instead, population status increased strongly with distance from the Central Valley, a pattern the authors note is consistent with the reported west-to-east spread of chytridiomycosis.

Knapp, R. A., Briggs, C. J., Smith, T. C., & Maurer, J. R. (2011). Nowhere to hide: impact of a temperature-sensitive amphibian pathogen along an elevation gradient in the temperate zone. *Ecosphere*, 2(8), art93. https://doi.org/10.1890/es11-00028.1

Knapp and colleagues tested the widely held idea that cold environments limit the temperature-sensitive chytrid fungus (*Batrachochytrium dendrobatidis*, Bd) and so provide amphibians a refuge, a hypothesis well supported in hot climates but largely untested in cold ones. Over five years in Yosemite National Park, they resurveyed all 285 Sierra Nevada yellow-legged frog (*Rana sierrae*) populations across an elevation and temperature gradient, described seasonal temperatures and corresponding Bd infection intensities, and translocated infected frogs from one source lake into five lakes along the gradient. Frog population size increased with elevation, but infection intensity and population persistence were unrelated to elevation, seasonal temperature had no significant effect on infection intensity, and post-translocation infection and survival were unrelated to water temperature. Contrary to the prevailing paradigm, the authors found little or no evidence that cold limits Bd even at the highest elevations, and conclude that high elevations in temperate mountains are unlikely to be refuges from Bd.

Berlow, E. L., Knapp, R. A., Ostoja, S. M., Williams, R. J., McKenny, H., Matchett, J. R., Guo, Q., Fellers, G. M., Kleeman, P., Brooks, M. L., & Joppa, L. (2013). A network extension of species occupancy models in a patchy environment applied to the Yosemite toad (*Anaxyrus canorus*). *PLoS ONE*, 8(8), e72200. https://doi.org/10.1371/journal.pone.0072200

Berlow and colleagues extended species occupancy modeling by combining patch-level predictions of habitat quality with network analysis of connectivity among patches, a method for identifying the sites most important to a rare species' regional persistence from limited survey data. They applied it to the Yosemite toad (*Anaxyrus canorus*), a meadow-breeding species then proposed for federal listing, across more than 2,500 meadows in Yosemite National Park. A generalized linear model correctly predicted about 84% of presence–absence records in a withheld test set, and each meadow's predicted quality was then iteratively boosted by the quality of neighboring meadows within dispersal distance. The resulting network distinguished linked clusters of high-quality meadows, isolated high-quality meadows, lower-quality meadows whose persistence may depend on their neighbors, and meadows with outsized influence on park-wide breeding patterns, providing a basis for designating core conservation areas.

Bradford, D. F., Stanley, K. A., Tallent, N. G., Sparling, D. W., Nash, M. S., Knapp, R. A., McConnell, L. L., & Massey Simonich, S. L. (2013). Temporal and spatial variation of atmospherically deposited organic contaminants at high elevation in Yosemite National Park, California, USA. *Environmental Toxicology and Chemistry*, 32(3), 517–525. https://doi.org/10.1002/etc.2094

This study sampled shallow-water sediments and Sierran treefrog (*Pseudacris sierra*) tadpoles at four high-elevation sites in Yosemite National Park twice each summer from 2006 to 2008, analyzing pesticides, polycyclic aromatic hydrocarbons (PAHs), and polychlorinated biphenyls. Concentrations of both historic- and current-use pesticides were generally lower than in Sequoia–Kings Canyon, consistent with a shorter air-transport path over farmland and less nearby pesticide use, whereas sediment PAHs were generally higher in Yosemite. Some DDT-related sediment concentrations in both parks exceeded guideline thresholds, but tadpole cholinesterase activity did not differ consistently between parks. Site, sampling period, and year all contributed significantly to variation in contaminant levels.

Deiner, K., Knapp, R. A., Boiano, D. M., & May, B. (2013). Increased accuracy of species lists developed for alpine lakes using morphology and cytochrome oxidase I for identification of specimens. *Molecular Ecology Resources*, 13(5), 820–831. https://doi.org/10.1111/1755-0998.12130

Deiner, Knapp, Boiano, and May compared benthic macroinvertebrate species lists for ten alpine lakes in Sequoia National Park built from morphology alone versus morphology combined with DNA barcoding of cytochrome oxidase I (COI). Of 77 unique taxa, 32 were reliably identified to species, and 20 of those were identified only through matches to the Barcode of Life reference library. The combined approach produced significantly higher within-lake richness at the order, family, genus, and species levels, and identified roughly 75% of damaged or immature specimens that morphology could not. The authors recommend barcoding, especially where morphological identification is insufficient.

Garwood, J. M., Knapp, R. A., Pope, K. L., Grasso, R. L., Magnuson, M. L., & Maurer, J. R. (2013). Use of historically fishless high-mountain lakes and streams by nearctic river otters (*Lontra canadensis*). *Northwestern Naturalist*, 94, 51–66.

Garwood and colleagues compiled records of North American river otters (*Lontra canadensis*) from high-elevation waters (above 1,100 m) in California, where fish and crayfish were historically absent. Applying evidentiary standards, they assembled 126 records showing that otters are widespread in the Klamath, southern Cascades, and Sierra Nevada, including above 3,000 m; 63% were verified and 37% unverified, with similar temporal and habitat patterns, and half dated from 1991 to 2010. About 93% of water bodies with otter records contained nonnative fish or crayfish, and where foraging was observed, otters ate nonnative prey in 89% of accounts and native frogs or invertebrates in 22%. The authors conclude that it is unclear whether otters used these waters before fish introductions and call for research on whether nonnative prey facilitated their expansion into high-mountain habitats.

Miner, B. E., Knapp, R. A., Colbourne, J. K., & Pfrender, M. E. (2013). Evolutionary history of alpine and subalpine *Daphnia* in western North America. *Freshwater Biology*, 58(7), 1512–1522. https://doi.org/10.1111/fwb.12152

Miner, Knapp, Colbourne, and Pfrender built a molecular phylogeny from two mitochondrial genes (ND5 and COI) for *Daphnia pulex*-complex populations in alpine and subalpine lakes and ponds of western North America, addressing phylogenetic uncertainty that had obscured the evolution of carapace melanism, often interpreted as an adaptation to ultraviolet radiation. The analysis confirmed a distinct mitochondrial clade for *Daphnia melanica*, which proved more geographically widespread than previously recognized and occurs at several disjunct sites. Melanism, once used as a diagnostic species character, varied within the *D. melanica* clade. The authors suggest this variation may reflect parallel, independent losses and gains of melanism.

Matchett, J. R., Stark, P. B., Ostoja, S. M., Knapp, R. A., McKenny, H. C., Brooks, M. L., Langford, W. T., Joppa, L. N., & Berlow, E. L. (2015). Detecting the influence of rare stressors on rare species in Yosemite National Park using a novel stratified permutation test. *Scientific Reports*, 5, 10702. https://doi.org/10.1038/srep10702

Matchett, Stark, Ostoja, Knapp, and colleagues addressed the difficulty of drawing causal inference from observational data on rare species exposed to rare stressors, a common problem in conservation decision-making. Their method first predicts occupancy from all predictors except the stressor, ranking habitat patches by suitability, and then applies a stratified permutation test of the stressor–species association within comparable strata. They applied it in Yosemite National Park to two pairs in which both stressor and species occur in roughly 10% of habitat patches: nonnative fish and the endangered Sierra Nevada yellow-legged frog (*Rana sierrae*), and packstock use and the threatened Yosemite toad (*Anaxyrus canorus*). The test confirmed the known negative effect of fish on *R. sierrae* but found no evidence that current packstock use affects toad breeding, and the authors suggest the approach is broadly applicable for extracting understanding, not just prediction, from observational data.

Zhou, H., Hanson, T., & Knapp, R. (2015). Marginal Bayesian nonparametric model for time to disease arrival of threatened amphibian populations. *Biometrics*, 71(4), 1101–1110. https://doi.org/10.1111/biom.12345

Zhou, Hanson, and Knapp developed a statistical model to predict when the amphibian chytrid fungus (*Batrachochytrium dendrobatidis*), which has driven hundreds of amphibian species to extinction, arrives in a host population. Because arrival-time data are strongly spatially structured and violate the proportional hazards assumption, they proposed a marginal Bayesian nonparametric survival model for spatially correlated right-

censored data, with log survival times following a mixture of normals under a linear-dependent Dirichlet process and a Gaussian copula capturing spatial dependence. A full-scale approximation makes large spatial correlation matrices tractable, and posterior computation uses an efficient MCMC algorithm with delayed rejection, implemented in the R package spBayesSurv. The model was evaluated by simulation and applied to threatened frog populations in Sequoia–Kings Canyon National Park.

Knapp, R. A., Fellers, G. M., Kleeman, P. M., Miller, D. A. W., Vredenburg, V. T., Rosenblum, E. B., & Briggs, C. J. (2016). Large-scale recovery of an endangered amphibian despite ongoing exposure to multiple stressors. *Proceedings of the National Academy of Sciences USA*, 113(42), 11889–11894. https://doi.org/10.1073/pnas.1600983113

Knapp and colleagues used more than 7,000 surveys conducted over 20 years across Yosemite National Park to assess whether the endangered Sierra Nevada yellow-legged frog (*Rana sierrae*) can recover while still exposed to introduced fish, chytridiomycosis, and pesticides. Frog abundance increased sevenfold, about 11% per year, with gains in hundreds of populations, a rare example of amphibian recovery at an ecologically meaningful scale. A laboratory experiment suggested that frogs had become less susceptible to chytridiomycosis, and removal or loss of nonnative fish after stocking ended also contributed. The authors conclude that where habitat is relatively intact and stressors are reduced by management or host adaptation, some amphibian declines may be at least partly reversible.

Piovia-Scott, J., Sadro, S., Knapp, R. A., Sickman, J., Pope, K. L., & Chandra, S. (2016). Variation in reciprocal subsidies between lakes and land: perspectives from the mountains of California. *Canadian Journal of Fisheries and Aquatic Sciences*, 73(11), 1691–1701. https://doi.org/10.1139/cjfas-2015-0549

Piovia-Scott and colleagues synthesized data from California mountain lake catchments to examine how reciprocal resource flows between lakes and land change along an elevational gradient and with the introduction of trout as a top aquatic predator. At low elevations, well-developed terrestrial vegetation delivers relatively large amounts of organic matter to lakes, whereas sparse vegetation at high elevations contributes little organic material but more inorganic nitrogen. Amphibians and emerging aquatic insects transport lake-derived resources back to land, and trout reduce these lake-to-land subsidies by consuming both. The resulting conceptual model proposes that in-lake processes increasingly influence terrestrial consumers at higher elevations, while terrestrial inputs drive lake processes at lower elevations.

Smith, T. C., Knapp, R. A., & Briggs, C. J. (2016). Declines and extinctions of mountain yellow-legged frogs have small effects on benthic macroinvertebrate communities. *Ecosphere*, 7(6), e01327. https://doi.org/10.1002/ecs2.1327

Smith and colleagues examined whether disease-driven declines and extinctions of mountain yellow-legged frogs (*Rana muscosa* and *R. sierrae*), formerly abundant and highly connected predators and grazers in Sierra Nevada lakes, altered benthic macroinvertebrate communities. Resurveys of 22 alpine lakes grouped by frog status (extant, declining, or extirpated) found taxonomic richness about 17% higher where frogs were declining or gone, but no strong multivariate community differences and no individual taxa differing in abundance. Reanalysis of snapshot survey data from 157 lakes likewise showed only small differences between lakes with and without frogs. The authors conclude that even the loss of a conspicuous, highly connected species can have small and subtle ecological effects.

Jani, A. J., Knapp, R. A., & Briggs, C. J. (2017). Epidemic and endemic pathogen dynamics correspond to distinct host population microbiomes at a landscape scale. *Proceedings of the Royal Society B: Biological Sciences*, 284(1857), 20170944. https://doi.org/10.1098/rspb.2017.0944

Jani, Knapp, and Briggs tested whether skin bacterial communities of *Rana sierrae* are associated with whether chytrid (Bd) infection leads to endemic coexistence or epidemic extirpation, using populations sampled before and after pathogen arrival to infer causation. Microbiome diversity and composition differed between endemic and epidemic populations, and composition was most strongly predicted by infection intensity, indicating that Bd shapes the microbiome. Bacterial richness, in contrast, was largely unaffected by infection but strongly predicted by geographic region, suggesting environmental or host-genetic influences. Higher richness was associated with population persistence and remained stable during infection, suggesting a possible, though untested, role in disease resistance.

Poorten, T. J., Knapp, R. A., & Rosenblum, E. B. (2017). Population genetic structure of the endangered Sierra Nevada yellow-legged frog (*Rana sierrae*) in Yosemite National Park based on multi-locus nuclear data from swab samples. *Conservation Genetics*, 18(4), 731–744. https://doi.org/10.1007/s10592-016-0923-5

Poorten and colleagues characterized the population genetic structure of the endangered Sierra Nevada yellow-legged frog (*Rana sierrae*) in Yosemite National Park using minimally invasive swabs from 23 sites, analyzing mitochondrial DNA and 43 transcriptome-derived nuclear markers. Mitochondrial data partly confirmed two previously reported haplotype groups, though only one had strong support as a single lineage, and denser sampling

showed the groups were not cleanly divided between the Merced and Tuolumne drainages. Nuclear data revealed isolation by distance and three clusters of genetically similar sites. The results can guide translocations, reintroductions, and monitoring, and show that swab archives collected for pathogen testing can also yield multilocus host genetic data.

Smith, T. C., Picco, A. M., & Knapp, R. (2017). Ranaviruses infect mountain yellow-legged frogs (*Rana sierrae*) threatened by *Batrachochytrium dendrobatidis*. *Herpetological Conservation and Biology*, 12, 149–159.

Smith, Picco, and Knapp investigated occasional tadpole mortality not associated with chytrid (*Batrachochytrium dendrobatidis*, Bd) epizootics, observed during 15 years of surveys of mountain yellow-legged frogs (*Rana muscosa* and *R. sierrae*) in Kings Canyon National Park. Tadpoles collected during die-offs in 2001 and 2005 from five lakes in one large basin (83 ponds and lakes in a 1,000-ha catchment) tested positive for ranavirus, but a 2006 survey of 17 populations within 25 km found ranavirus in only 6 of 174 tadpoles, all from the same basin, five of them coinfected with Bd. Population and Bd data from that basin for 1997–2009 showed that, despite ranavirus epizootics and tadpole mortality, frog populations did not decline until Bd epizootics caused massive adult mortality. The authors conclude that ranavirus could reduce recruitment but has likely not contributed to mountain yellow-legged frog declines compared with the severe impact of Bd.

Wilber, M. Q., Knapp, R. A., Toothman, M., & Briggs, C. J. (2017). Resistance, tolerance and environmental transmission dynamics determine host extinction risk in a load-dependent amphibian disease. *Ecology Letters*, 20(9), 1169–1181. https://doi.org/10.1111/ele.12814

Wilber, Knapp, Toothman, and Briggs combined models and experiments on the mountain yellow-legged frog, which has suffered local extirpations from *Batrachochytrium dendrobatidis* (Bd), to assess the relative importance of transmission, host resistance, and host tolerance for disease-induced extinction. Although such extinctions are generally considered rare, several emerging pathogens with load-dependent pathology have caused them. Transmission from an environmental Bd reservoir increased both pathogen invasion probability and host extinction risk, but extinction dynamics were far more sensitive to host resistance and tolerance than to transmission. The authors argue that this pattern should hold generally for load-dependent pathogens, where nonlinear resistance and tolerance functions can interact so that small changes produce large shifts in extinction risk.

Joseph, M. B., & Knapp, R. A. (2018). Disease and climate effects on individuals drive post-reintroduction population dynamics of an endangered amphibian. *Ecosphere*, 9(11), e02499. https://doi.org/10.1002/ecs2.2499

Joseph and Knapp used long-term mark–recapture data from reintroduced populations of the endangered Sierra Nevada yellow-legged frog (*Rana sierrae*) to quantify how the amphibian chytrid fungus (*Batrachochytrium dendrobatidis*) affects population dynamics after reintroduction. High infection intensities were associated with higher detectability and lower survival of individual frogs. When mean infection intensity in a population was high, adults were more likely to become infected and less likely to clear infection, and heavily infected frogs died at higher rates. The authors argue for probabilistic methods that propagate uncertainty in infection intensity from imperfect field data to better understand disease impacts and guide conservation actions.

Byrne, A. Q., Vredenburg, V. T., Martel, A., Pasmans, F., Bell, R. C., Blackburn, D. C., Bletz, M. C., Bosch, J., Briggs, C. J., Brown, R. M., Catenazzi, A., Familiar López, M., Figueroa-Valenzuela, R., Ghose, S. L., Jaeger, J. R., Jani, A. J., Jirku, M., Knapp, R. A., Muñoz, A., Portik, D. M., Richards-Zawacki, C. L., Rockney, H., Rovito, S. M., Stark, T., Sulaeman, H., Tao, N. T., Voyles, J., Waddle, A. W., Yuan, Z., & Rosenblum, E. B. (2019). Cryptic diversity of a widespread global pathogen reveals expanded threats to amphibian conservation. *Proceedings of the National Academy of Sciences*, 116(41), 20382–20387. https://doi.org/10.1073/pnas.1908289116

Byrne and colleagues developed a custom genotyping assay that works on degraded DNA from non-invasive skin swabs and applied it to a global collection of *Batrachochytrium dendrobatidis* (Bd) samples. They discovered a new, highly divergent lineage, Bd-ASIA-3, that appears widespread in Southeast Asia and co-occurs with the global panzootic lineage (Bd-GPL) at several sites. The work also refined the known global distribution of Bd-GPL and expanded the documented range of the Bd-CAPE lineage. The authors call for monitoring where Bd lineages come into contact and in regions with poorly characterized lineage diversity, noting that archived swabs make this inexpensive and allow reconstruction of the pathogen's history.

Ellison, S., Knapp, R. A., Sparagon, W., Swei, A., & Vredenburg, V. T. (2019). Reduced skin bacterial diversity correlates with increased pathogen infection intensity in an endangered amphibian host. *Molecular Ecology*, 28(1), 127–140. https://doi.org/10.1111/mec.14964

Ellison and colleagues examined skin bacterial communities in two Sierra Nevada yellow-legged frog (*Rana sierrae*) populations that likely survived past chytrid (*Batrachochytrium dendrobatidis*, Bd) outbreaks and now persist with stable, lower-level infection. Using 16S

rRNA metabarcoding, they found that heavily infected juveniles had markedly lower bacterial richness and evenness and much less among-individual variation than lightly infected juveniles and adults. More than 90% of sequences from heavily infected frogs belonged to the order Burkholderiales, compared with 54% in lightly infected frogs. All bacterial metabolites tested in the laboratory inhibited Bd growth, and the authors conclude that Bd disrupts skin microbiome structure while these bacteria may inform anti-Bd probiotic development.

Rothstein, A. P., Knapp, R. A., Bradburd, G. S., Boiano, D. M., Briggs, C. J., & Rosenblum, E. B. (2020). Stepping into the past to conserve the future: Archived skin swabs from extant and extirpated populations inform genetic management of an endangered amphibian. *Molecular Ecology*, 29(14), 2598–2611. https://doi.org/10.1111/mec.15515

Rothstein and colleagues used 598 archived skin swabs, originally collected for pathogen testing, from extant and extirpated mountain yellow-legged frog (*Rana sierrae*/*R. muscosa*) populations across 48 lake basins in Sequoia and Kings Canyon National Parks to build an Illumina-based nuclear amplicon dataset. Disease-driven declines and limited knowledge of population genetic structure had complicated long-term planning for translocations and reintroductions. Samples fell into three main genetic clusters matching watershed boundaries, with evidence of historical gene flow across watersheds along a north–south axis. Genetic diversity did not differ significantly among populations of different disease status, and the authors present the work as a framework for using minimally invasive samples in genetic management of threatened species.

Ellison, S., Knapp, R., & Vredenburg, V. (2021). Longitudinal patterns in the skin microbiome of wild, individually marked frogs from the Sierra Nevada, California. *ISME Communications*, 1(1), 45. https://doi.org/10.1038/s43705-021-00047-7

Ellison and colleagues tracked the skin bacterial microbiome of 19 individually marked adult Sierra Nevada yellow-legged frogs (*Rana sierrae*) in two wild populations over two years. Frogs with low bacterial diversity, dominated by the order Burkholderiales, had more stable communities than frogs with higher diversity. Highly abundant bacterial taxa (amplicon sequence variants) persisted more consistently than rare ones, and those at intermediate abundance changed most over time. The authors suggest that probiotic strategies against the chytrid fungus should target bacteria that reach high abundance on some frogs, as these are most likely to persist.

Jani, A. J., Bushell, J., Arisdakessian, C. G., Belcaid, M., Boiano, D. M., Brown, C., & Knapp, R. A. (2021). The amphibian microbiome exhibits poor resilience following pathogen-induced disturbance. *The ISME Journal*, 15, 1628–1640. https://doi.org/10.1038/s41396-020-00875-w

Jani and colleagues assessed how well the amphibian skin microbiome recovers from disturbance by *Batrachochytrium dendrobatidis* (Bd), tracking laboratory-reared mountain yellow-legged frogs before infection, during infection, and after Bd was cleared with an antifungal drug. Bd infection disrupted microbiome composition and shifted the relative abundances of several dominant bacterial taxa. Clearing the infection did not restore community composition (measured as UniFrac distance) or the abundances of dominant groups, indicating either an alternate stable state or extremely slow recovery; in either case, resilience was low. Antifungal treatment and Bd clearance also reduced among-frog variability in microbial communities, which the authors attribute to similar taxa colonizing the vacancies left by Bd's removal, with possible implications for conservation of this endangered frog.

Joseph, M. B., & Knapp, R. A. (2021). Using visual encounter data to improve capture–recapture abundance estimates. *Ecosphere*, 12(2), e03370. https://doi.org/10.1002/ecs2.3370

Joseph and Knapp showed that visual encounters of animals that are seen but not captured, and therefore not identified, can improve capture–recapture abundance estimates. If each animal is encountered at most once per survey, these failed captures provide a lower bound on population size that can increase precision, with the greatest benefit when capture probabilities are low and surveys are few. The authors present a hierarchical Bayesian model that integrates failed captures and other auxiliary encounter counts with standard capture–recapture models. The approach is aimed especially at hard-to-capture species in data-limited settings, such as many amphibians.

Rothstein, A. P., Byrne, A. Q., Knapp, R. A., Briggs, C. J., Voyles, J., Richards-Zawacki, C. L., & Rosenblum, E. B. (2021). Divergent regional evolutionary histories of a devastating global amphibian pathogen. *Proceedings of the Royal Society B: Biological Sciences*, 288(1953), 20210782. https://doi.org/10.1098/rspb.2021.0782

Rothstein and colleagues used amplicon sequencing of non-invasive skin swabs to reconstruct the history of the Global Panzootic Lineage of the amphibian chytrid fungus (BdGPL) in two regions with well-documented epizootics: the Sierra Nevada of California and central Panama. The working hypothesis was that BdGPL had been recently introduced to both regions and spread rapidly as a wave. Genetic data confirmed recent, rapid spread in Panama, but in the Sierra Nevada BdGPL showed strong spatial structure, high genetic diversity, and an older history in time-dated phylogenies. The authors conclude that similar epizootic signatures can arise from very different evolutionary

histories, that BdGPL may have been present in some regions longer than assumed, and that die-off observations should be combined with genetic data to reconstruct outbreaks accurately.

Tiberti, R., Buchaca, T., Boiano, D., Knapp, R. A., Pou Rovira, Q., Tavecchia, G., Ventura, M., & Tenan, S. (2021). Alien fish eradication from high mountain lakes by multiple removal methods: Estimating residual abundance and eradication probability in open populations. *Journal of Applied Ecology*, 58(5), 1055–1068. https://doi.org/10.1111/1365-2664.13857

Tiberti and colleagues developed an analytical approach for estimating residual population size and eradication probability after mechanical removal of invasive fish, extending removal models to open populations (allowing survival and recruitment during eradication) and to multiple concurrent removal methods (electrofishing, gill netting, and fyke netting) across sites and seasons. They applied the model to 20 projects in Europe and the United States aimed at restoring high mountain lakes to fishless conditions. In control lakes where eradication had been confirmed by more than five years of post-removal surveys, the model indicated success, defined as a 95% credible interval for residual population size including fewer than two individuals, in 13 of 15 lake-by-species cases, and assigned an eradication probability of zero to projects still in progress. The approach can help managers estimate residual populations and optimize long-term eradication efforts.

Knapp, R. A., Joseph, M. B., Smith, T. C., Hegeman, E. E., Vredenburg, V. T., Erdman Jr, J. E., Boiano, D. M., Jani, A. J., & Briggs, C. J. (2022). Effectiveness of antifungal treatments during chytridiomycosis epizootics in populations of an endangered frog. *PeerJ*, 10, e12712. https://doi.org/10.7717/peerj.12712

Knapp and colleagues tested three antifungal interventions in wild populations of the endangered mountain yellow-legged frog during or just after chytrid (Bd) die-offs: itraconazole treatment of early life stages, itraconazole treatment of adults, and augmentation of subadults with the protective skin bacterium *Janthinobacterium lividum*. Itraconazole rapidly reduced Bd loads, but in early life stages loads and subadult survival returned to pre-treatment levels within a year with no effect on population persistence. In adults, treatment reduced Bd loads and increased survival for the full three-year follow-up, suggesting an adaptive immune response, yet poor recruitment led the population to decline to near-extirpation. Bacterial augmentation raised *J. lividum* levels only briefly and gave no protection, and the authors attribute the overall lack of lasting benefit mainly to the inability of early life stages to mount an effective immune response.

Smith, T. C., Knapp, R. A., Imperato, J., Miller, K., & Rose, D. (2022). *Rana sierrae* (Sierra Nevada Yellow-legged Frog). Behavior and Diet. *Herpetological Review*, 53(3), 478–479.

No abstract (natural history note). This *Herpetological Review* note reports observations of behavior and diet in the Sierra Nevada yellow-legged frog (*Rana sierrae*).

Wilber, M. Q., Knapp, R. A., Smith, T. C., & Briggs, C. J. (2022). Host density has limited effects on pathogen invasion, disease-induced declines and within-host infection dynamics across a landscape of disease. *Journal of Animal Ecology*, 91(12), 2451–2464. https://doi.org/10.1111/1365-2656.13823

Wilber, Knapp, Smith, and Briggs tested the widely held expectation that host density strongly shapes wildlife population responses to pathogen invasion, using hundreds of mountain yellow-legged frog populations whose densities spanned orders of magnitude. With a hidden Markov model accounting for imperfectly observed outbreak states, they examined whether Bd invasion led to outbreaks, the magnitude of subsequent declines, and within-host infection dynamics. Contrary to predictions from small-scale experiments, host density had minimal effects on all three outcomes, whereas summer temperature, winter severity, and the presence of pathogen reservoirs were better predictors. The authors conclude that small-scale transmission experiments may not scale to field outcomes and that density–disease feedbacks may not determine reintroduction success in Bd-affected species.

Byrne, A. Q., Rothstein, A. P., Smith, L. L., Kania, H., Knapp, R. A., Boiano, D. M., Briggs, C. J., Backlin, A. R., Fisher, R. N., & Rosenblum, E. B. (2024). Revisiting conservation units for the endangered mountain yellow-legged frog species complex (*Rana muscosa*, *Rana sierrae*) using multiple genomic methods. *Conservation Genetics*, 25, 591–606. https://doi.org/10.1007/s10592-023-01568-5

This study re-evaluated conservation units for the endangered mountain yellow-legged frogs (*Rana muscosa* and *R. sierrae*), whose management has relied mainly on a single mitochondrial marker. Using amplicon sequencing and exome capture on buccal swabs, museum tissues, and archived skin swabs from across the range, the authors identified five major genetic clusters and largely confirmed the previously proposed species boundary in Kings Canyon National Park, with a few site-level exceptions. *R. muscosa* clustered sharply, whereas *R. sierrae* showed admixture consistent with stepping-stone population structure. The southern *R. muscosa* cluster had the lowest heterozygosity and long runs of homozygosity, reflecting severe declines, and the results clarify management-unit designations.

Dodge, C. M., Brown, C., Lind, A. J., Knapp, R. A., Wilkinson, L. R., & Vredenburg, V. T. (2024). Historical and contemporary impacts of an invasive fungal pathogen on the Yosemite toad. *Biological Conservation*, 291, 110504. https://doi.org/10.1016/j.biocon.2024.110504

Dodge and colleagues examined the role of the amphibian chytrid fungus (*Batrachochytrium dendrobatidis*, Bd) in the decline of the Yosemite toad (*Anaxyrus canorus*), which suffered mass die-offs in the late 1970s, two decades before Bd was described. They tested 719 museum specimens collected from 1915 to 2005 and sampled 1,678 live toads from 2004 to 2012, modeling biological and environmental predictors of infection. Bd appeared in the toad's range at about the time of the historical declines and is now widespread, with infection associated with life stage, elevation, and precipitation; juveniles had the highest prevalence, sometimes with loads above levels lethal in other species. The authors conclude that Bd likely played a significant, previously unrecognized role in the decline and may still depress survival and recruitment, with implications for many species that declined before Bd was discovered.

Knapp, R. A., Wilber, M. Q., Joseph, M. B., Smith, T. C., & Grasso, R. L. (2024). Reintroduction of resistant frogs facilitates landscape-scale recovery in the presence of a lethal fungal disease. *Nature Communications*, 15, 9436. https://doi.org/10.1038/s41467-024-53608-4

Knapp and colleagues report a 15-year effort to reintroduce Sierra Nevada yellow-legged frogs (*Rana sierrae*) from populations that had recovered from chytrid (Bd) epizootics and developed resistance into vacant habitats where Bd remained present. Reintroduced frogs established new populations despite Bd, with success depending on site, cohort, and individual characteristics. Population viability modeling indicated that many reintroduced populations have a low probability of extinction over the next 50 years. The authors conclude that reintroducing resistant individuals can drive landscape-scale recovery of species impacted by novel pathogens, an approach applicable to other amphibians and taxa.

Schrock, S. A. R., Walsman, J. C., DeMarchi, J., LeSage, E. H., Ohmer, M. E. B., Rollins-Smith, L. A., Briggs, C. J., Richards-Zawacki, C. L., Woodhams, D. C., Knapp, R. A., Smith, T. C., Haddad, C. F. B., Becker, C. G., Johnson, P. T. J., & Wilber, M. Q. (2025). Do fungi look like macroparasites? Quantifying the patterns and mechanisms of aggregation for host–fungal parasite relationships. *Proceedings of the Royal Society B: Biological Sciences*, 292(2043), 20242013. https://doi.org/10.1098/rspb.2024.2013

Schrock and colleagues asked whether infection intensities of the amphibian chytrid fungus (*Batrachochytrium dendrobatidis*, Bd) are aggregated among hosts in the way macroparasites are, where a few hosts carry most parasites, and whether aggregation reflects underlying biological processes. Using the slope of Taylor's power law across amphibian–Bd systems, they found Bd intensity distributions to be more aggregated than those of many macroparasites and closely approximated by lognormal distributions. Aggregation patterns tracked epizootic phase (invasion, post-invasion, or enzootic) and intensity-dependent mortality, and intensity-dependent models indicated shifts in aggregation consistent with evolved host resistance where Bd persists after declines. The authors conclude that Bd aggregation is highly conserved and carries signatures of biological processes, offering a basis for a common theoretical framework for host–fungal dynamics.

Byrne, A. Q., Rothstein, A. P., Knapp, R. A., & Rosenblum, E. B. (2026). Genomic correlates of disease recovery in natural populations of the Sierra Nevada yellow-legged frog (*Rana sierrae*). *Molecular Ecology*, 36, e70348. https://doi.org/10.1111/mec.70348

Byrne, Rothstein, Knapp, and Rosenblum tested whether evolution contributed to the recovery of Sierra Nevada yellow-legged frog (*Rana sierrae*) populations after severe declines caused by the chytrid fungus *Batrachochytrium dendrobatidis*, a role that had previously been supported only by circumstantial evidence. They compared genomes of frogs from nearby disease-naive and recovering populations and used multiple genome scans to detect signatures of selection. Recovery was associated with outlier variants in genes related to skin integrity and intracellular regulation, an interferon-related gene, and RIN3, a gene that may function in disease defense and wound healing, while genetic diversity did not differ between naive and recovering populations. The study provides a rare natural-population example suggesting that selection on adaptive alleles can enable recovery from a novel pathogen.

Land, P. R., Lattig, G. Q., Walderich, L. M., Staines, J. W., Peri, F. D., & Knapp, R. A. (2026). *Hydromantes platycephalus*. (Mount Lyell Salamander). Lentic habitat use. *Herpetological Review*, in press.

No abstract (natural history note). This *Herpetological Review* note documents use of lentic (lake or pond) habitat by the Mount Lyell salamander (*Hydromantes platycephalus*), a species typically associated with rocky terrestrial habitats. The note is in press.